

On the optimal reconstruction and control of adaptive optical systems with mirror dynamics

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Received September 17, 2009; revised November 26, 2009; accepted December 4, 2009;
posted December 14, 2009 (Doc. ID 114950); published January 29, 2010

In adaptive optics (AO) the deformable mirror (DM) dynamics are usually neglected because, in general, the DM can be considered infinitely fast. Such assumption may no longer apply for the upcoming Extremely Large Telescopes (ELTs) with DM that are several meters in diameter with slow and/or resonant responses. For such systems an important challenge is to design an optimal regulator minimizing the variance of the residual phase. In this contribution, the general optimal minimum-variance (MV) solution to the full dynamical reconstruction and control problem of AO systems (AOSs) is established. It can be looked upon as the parent solution from which simpler (used hitherto) suboptimal solutions can be derived as special cases. These include either partial DM-dynamics-free solutions or solutions derived from the static minimum-variance reconstruction (where both atmospheric disturbance and DM dynamics are neglected altogether). Based on a continuous stochastic model of the disturbance, a state-space approach is developed that yields a fully optimal MV solution in the form of a discrete-time linear-quadratic-Gaussian (LQG) regulator design. From this LQG standpoint, the control-oriented state-space model allows one to (1) derive the optimal state-feedback linear regulator and (2) evaluate the performance of both the optimal and the sub-optimal solutions. Performance results are given for weakly damped second-order oscillatory DMs with large-amplitude resonant responses, in conditions representative of an ELT AO system. The highly energetic optical disturbance caused on the tip/tilt (TT) modes by the wind buffeting is considered. Results show that resonant responses are correctly handled with the MV regulator developed here. The use of sub-optimal regulators results in prohibitive performance losses in terms of residual variance; in addition, the closed-loop system may become unstable for resonant frequencies in the range of interest. © 2010 Optical Society of America

OCIS codes: 010.1080, 100.3190, 230.4040.

1. OVERVIEW OF LINEAR-QUADRATIC-GAUSSIAN METHODS IN ASTRONOMICAL ADAPTIVE OPTICS

Atmospheric turbulence induces high-frequency erratic fluctuations in both intensity and phase of incoming light wavefronts. The latter severely affects astronomical imaging devices, which integrate the flow of light over long exposure times. This results in a particularly negative blurring effect, which unlike usual instrument-induced distortions can be corrected for only to a limited extent by image post-processing [1].

The obvious alternative is to seek to counter the effects of turbulence during the very process of image formation. This can be achieved by deploying an adaptive optics (AO) system, whose basic set-up is as follows: light entering the telescope is made to bounce on a deformable mirror (DM), the shape of which is adjusted in real time using measurements of the resulting corrected wavefront provided by a wavefront sensor (WFS). In this configuration the system is said to operate in closed-loop operation (Fig. 1 below) in opposition to the open-loop operation where measurements are directly made out of the uncorrected disturbance.

Minimum-variance (MV) wavefront estimation and control in AO is part of several research efforts, for it maximizes the well-known Strehl ratio (SR) [2]. Several

partial strategies address the problem in different contexts with approximations and simplifications that allow for considerable reduction in complexity and therefore for ease of problem formulation. These include neglecting either the DM temporal dynamics or the phase evolution or even both altogether.

With the advent of the Very Large Telescopes (8–10 m class) and the forthcoming Extremely Large Telescopes (ELTs), whose primary mirror sizes go up to 42 m diameter and in general of adaptive telescopes (using AO in the optical path), current designs include DMs with large physical dimensions (>1 m diameter). Temporal dynamics of such DMs may require the development of dedicated analysis tools and synthesis of optimal regulators. For more information on these topics refer to the SPIE mirror's session and articles therein [3]. It is underlined that optimality refers to a given criterion and the models that are used.

The AO minimization problem is described by a quadratic criterion. With disturbances assumed normally distributed and the problem being linear, the linear-quadratic-Gaussian (LQG) formalism is a natural way to synthesize MV optimal controllers. It enables one to integrate the temporal dynamics of the phase by means of a linear and time-invariant (LTI) stochastic model and to optimally (in the MV sense) estimate the open-loop phase

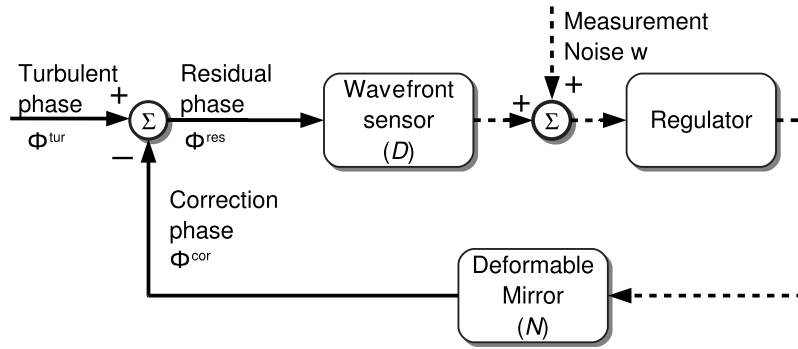


Fig. 1. General scheme of AO feedback control loop (closed-loop). Measurements are assumed to be affected by additive uncorrelated Gaussian zero-mean white noise w . Continuous/dashed lines for continuous/discrete-time variables.

from closed-loop present and past measurements. Following the pioneering contribution of Paschal *et al.* [4], Le Roux *et al.* [5] provided a discrete-time formulation and Kulcsár *et al.* [6] demonstrated equivalency of discrete- and continuous-time criteria. These contributions relied on a decomposition of the turbulence over a Zernike basis. Alternatively, a Fourier basis was used in [7]. Because Fourier modes can be assumed to be statistically independent under Taylor frozen flow hypothesis, this results in a decoupled LQG controller structure and thus in a considerable reduction of real-time computational burden.

The problem of determining a discrete-time regulator that minimizes a continuous-time AO performance criterion has been investigated by Looze [8,9]. Starting from continuous-time LTI models of both the turbulence and the DM, and when a quadratic penalty on the control energy is added to the performance criterion, Looze has shown that then the optimal controller can be obtained as the solution of a standard discrete-time LQG control problem. This formulation also allows one to optimally take into account asynchronous measurements. However, because of the presence of the additional control energy penalty, the resulting discrete-time LQG problem is not optimal with respect to the original continuous-time MV criterion (average residual phase variance). In addition, the starting point for the construction is to directly look for an LTI controller in output-feedback form. This results in a state vector whose physical sense is difficult to apprehend. As a consequence, it is also not very easy to factor out the effect of measurement delays and noise on achievable performance—a key issue in AO control design.

The approach presented here, which extends preliminary results in [10] to a broad class of LTI DM second-order dynamic models, builds on the model of Le Roux *et al.* [5]. It enables construction, under mild assumptions on the DM's dynamics, of a tractable discrete-time criterion that is equivalent to the continuous-time residual phase variance. In addition, this construction separates the MV optimization into a complete-state-information (CSI) optimal control problem involving weighted temporal averages of future values of the turbulence and the optimal MV prediction problem of estimating these turbulence values. The corresponding decomposition of the error budget enables evaluation of the impact of measurement delays and noise. Moreover, in the infinitely fast case, the complete-state-information control corresponds to a simple orthogonal projection of the optimal prediction of the turbulence average onto the DM space.

The solution to the full dynamical problem establishes the formalism to dynamically reconstruct the wavefront and control AO systems with disturbance and DM dynamics while minimizing the residual wavefront variance over the field of view. This solution can be looked upon as the parent from which simpler sub-optimal solutions are derived as special cases. It stands upon the LQG approach that splits up the design into two steps: (i) estimate the wavefront from noisy measurements and (ii) compute the DM commands that optimally compensate for the estimated value of the turbulence. In a linear Gaussian framework, the separation principle guarantees global optimality of this two-step scheme [11,12].

Making use of a continuous stochastic model of the disturbance, a state-space model is constructed that yields a fully optimal MV solution in the form of a discrete-time LQG control. It further provides a means to (a) determine the optimal feedback regulator and (b) assess the performance of any state-feedback linear discrete regulators. Such regulators will be derived in a particularly instructive scalar case with a resonant second-order DM in an ELT scenario, where large-amplitude wind-induced disturbances are considered.

The remainder of this paper is organized as follows. In this section the optimality criterion is recalled, and in Section 2 the fundamentals of the LQG approach are reviewed, detailing how the AO control problem inscribes in it. For readiness of comprehension and explanatory purposes, the LQG solution neglecting any DM dynamics is presented in Section 3 and later extended in Section 4 to the full-frame case with DM dynamics. From the solution found, in Subsection 4.C the means to evaluate the MV criterion are determined along with the state space employed in the derivation of the optimal regulator. With the bulk of the theory in hand, in Section 5 several examples of ELT tip/tilt control are illustrated through analytical and numerical simulations, after discussing the construction of the disturbance stochastic and DM deterministic LTI models.

A. Notation

Throughout this paper, the value $u(t)$ of a continuous variable u at sampling instants $t = kT_s$ will be denoted as $u_k \triangleq u(kT_s)$ and average values over the integration interval T_s by $\bar{u}_k \triangleq 1/T_s \int_{(k-1)T_s}^{kT_s} u(t)dt$. In the general case, x refers to a state vector with multiple entries. Matrices A_d , B_d , C_d , and D_d form the standard state-space system with

state disturbance and measurement noise of known variance Σ_v and Σ_w respectively. Measurements are concatenated in y and commands in u . Matrices D and N represent the wavefront sensor phase-to-measurements and the DM voltage-to-phase linear operations. Continuous J^c and discrete J^d criteria use the whole or part of the state x , with weighting matrices (Q, S, R) . Superscripts c and d are for continuous and discrete versions of the variables/operators they are attached to, whereas tur and m stand, respectively, for turbulent disturbance (whether of atmospheric origin or other) and mirror. They are attached to the internal states of one or another model within the complete state-space model.

B. Optimality Criteria

In what follows, the minimization of the pupil-integrated mean square residual phase after AO correction is considered. Minimizing the variance of ϕ^{res} results in the maximization of the SR [2], leading to the continuous-time criterion,

$$\begin{aligned} J^c(u) &\triangleq \lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{res}(t)\|^2 dt \\ &= \lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{tur}(t) - \phi^{cor}(t)\|^2 dt, \end{aligned} \quad (1)$$

where the residual phase is the difference between the turbulent and correction phases, $\phi^{res} = \phi^{tur} - \phi^{cor}$. Refer to Fig. 1 for further details on the feedback control system.

From a control point of view, the objective of Eq. (1) defines a disturbance rejection problem, since the control u cannot act on the disturbance phase ϕ^{tur} .

Define $J^c(u)_k$ as the average value of the residual phase variance between sampling instants $t = kT_s$ and $t = (k+1)T_s$:

$$\begin{aligned} J^c(u)_k &\triangleq \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{res}(t)\|^2 dt \\ &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{tur}(t) - \phi^{cor}(t)\|^2 dt. \end{aligned} \quad (2)$$

With this notation, the continuous-time criterion of Eq. (1) can be rewritten as an infinite-horizon average over all temporal frames of duration T_s :

$$J^c(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^M J^c(u)_k. \quad (3)$$

Note also that by construction $J^c(u)_k$ depends only on control decisions made up to time $t = kT_s$, in other words on the discrete sequence $u_0, \dots, u_k$.

Under the standard assumption that ϕ^{tur} is a wide-sense-stationary stochastic process, and when a stabilizing linear time-invariant controller is used to generate the control u , it is immediately shown that the stochastic process $J^c(u)_{k \in \mathbb{N}}$, converges toward a stationary and ergodic process. The criterion $J^c(u)$ in Eq. (1) is thus almost surely (a.s.) equal to the mathematical expectation of $J^c(u)_k$:

$$J^c(u) \stackrel{a.s.}{=} E(J^c(u)_k). \quad (4)$$

2. REVIEW OF THE LQG CONTROL APPROACH

The LQ and LQG approaches are two standard design tools in optimal control theory [11,13]. Their main aspects are recalled here to illustrate how the inherent state-space formalism applies to the AO MV problem, Eqs. (1) and (2).

The state-space formalism consists of mathematically modeling a physical system as a set of inputs, outputs, and state variables related by first-order difference equations (discrete case). States are considered as column vectors, and the algebraic and difference equations are written in matrix form. With respect to the classical single-input single-output (SISO) systems for which frequency analysis is often preferred (in the form of transfer functions) using state spaces is a very convenient way to represent multiple-input multiple-output (MIMO) systems that combine variables evolving in several physically distinct spaces together with real-time computational delays. In an AO context, phase (and its temporal dynamics) along with the WFS measurements and DM controls can be explicitly included in the model.

Regulator synthesis in AO was hitherto based upon modal decoupling (see for example [14]). In so doing, one employs as many scalar regulators as controlled modes. Frequency analysis is therefore very handy. The commonly used integrators with gain (proportional-integral action) can be straightforwardly embedded in a more general state-space approach, which brings to light their internally hidden and unstable phase temporal model [6]. Though simple to synthesize, attainable performance is compromised.

The initial LQG setup consists of three essential requirements: (1) the system is linear, (2) the criterion is quadratic, and (3) the disturbances are zero mean, white, and Gaussian distributed. Consider the following general discrete-time linear system of the order n , $\forall k \geq 0$:

$$\begin{cases} x_{k+1} = \mathcal{A}_d x_k + \mathcal{B}_d u_k + \Gamma v_k \\ z_k = \mathcal{C}_d x_k + \mathcal{D}_d u_k + w_k \end{cases} \quad (5)$$

The variables v and w are assumed to be Gaussian white noises, i.e., sequences of independent and identically distributed vector-valued Gaussian variables, with distributions $v_k \sim \mathcal{N}(0, \Sigma_v^2)$ and $w_k \sim \mathcal{N}(0, \Sigma_w^2)$. For the sake of simplicity, it will also be assumed that for any choice of k, k' , state and measurement noises are independent.

In Eq. (5) the notion of state gathers both the disturbance and mirror states. The general system of Eq. (5) defines

1. a state evolution equation, where the state temporal dynamics are characterized by a state transition matrix $\mathcal{A}_d$ and an input vector $\mathcal{B}_d$ that brings about the effect of the control decisions u_k at instants $t = kT_s$ and state noise v_k . This latter component will be used to account for the random behavior of disturbance;

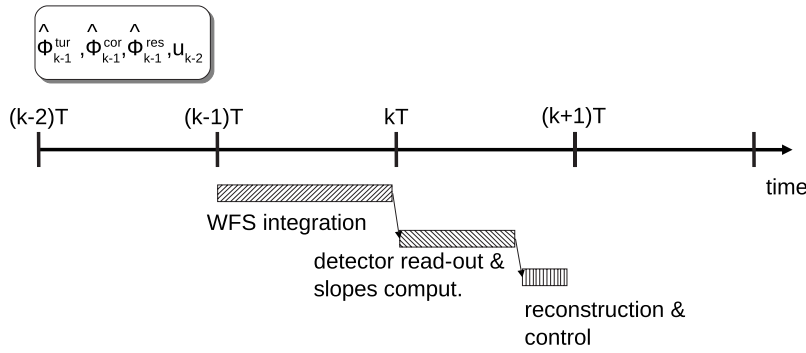


Fig. 2. Simplified temporal diagram of operations in AO. Controls u_k are applied at $t = kT_s$ over the interval $t \in [kT_s, (k+1)T_s]$. Any other sources of delay are neglected in order to have a total delay of two frames. Example: light is integrated from $t \in [(k-2)T_s, (k-1)T_s]$. From $t = (k-1)T_s$ onward the detector is read-out and measurements computed. Wavefront reconstruction and control take place together in the same time slot at the end of which the controls u_k are ready for application and kept constant by a zero-order hold through $t \in [kT_s, (k+1)T_s]$. The system is said to have a two-frame delay.

2. a measurement equation indicating which information is available at $t = kT_s$ for computing the control u_k . The measurement z_k is assumed to be the sum of a linear function of the internal state x_k corrupted with additive zero-mean Gaussian white noise $w_k \sim \mathcal{N}(0, \Sigma_w)$ independent of v_k (and hence of x_k).

Within a LQ/LQG framework, the problem at hand can be phrased as the determination of the control decisions $u_k \forall k > 0$ that minimize a quadratic criterion in the form

$$\mathcal{J}^d(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} (x_k^T Q x_k + u_k^T R u_k + 2x_k^T S u_k), \quad (6)$$

where Q , S , and R are weighting matrices to be obtained from the development of the AO MV criterion. To define a well-posed MV problem, it is necessary that Eq. (6) correspond to a proper quadratic form, i.e.,

$$P \triangleq \begin{pmatrix} Q & S \\ S^T & R \end{pmatrix} \geq 0. \quad (7)$$

This positiveness condition is in turn equivalent to $Q = Q^T$, $Q - SR^{-1}S^T \geq 0$, S , and $R = R^T > 0$.

Suppose that the state $x_k \forall k > 0$ is perfectly known. The solution to this CSI problem can be expressed in state feedback form as

$$u_k = -\mathcal{K}^{\text{opt}} x_k, \quad (8)$$

where $\mathcal{K}^{\text{opt}}$ is found from the solution of a control Riccati equation that involves the matrices Q , R , and S in Eq. (6).

In the general and more realistic incomplete state information (ISI) case the separation principle [11,13] applies: the optimal control has the same form as Eq. (8), where the state is replaced by its conditional expectation, denoted $\hat{x}$. Such state is estimated from the noisy and possibly delayed measurements; see the temporal arrangement of events in Fig. 2.

In other words, the optimal control decisions are retrieved from

$$u_k = -\mathcal{K}^{\text{opt}} \hat{x}_{k|k-1}, \quad (9)$$

using this estimate as if it were the true state in the state-feedback control. The subscript in the form $(k|k-1)$ means the conditional expectation of x_k with respect

to statistical priors and measurements $\{z_0, \dots, z_{k-1}\}$.

Provided that one has a linear model describing the state's dynamics, $\hat{x}_{k|k-1}$ is obtained recursively from the output of a Kalman filter with gain $\mathcal{L}^{\text{opt}}$ [13]. The state estimate is obtained from the recursive equation

$$\hat{x}_{k|k-1} = \mathcal{A}_d \hat{x}_{k-1|k-2} + \mathcal{B}_d u_{k-1} + \mathcal{L}^{\text{opt}}(z_{k-1} - \hat{z}_{k-1|k-2}), \quad (10)$$

where the gain

$$\mathcal{L}^{\text{opt}} = \mathcal{A}_d \Sigma \mathcal{C}_d^T (\mathcal{C}_d \Sigma \mathcal{C}_d^T + \Sigma_w)^{-1} \quad (11)$$

is obtained from

$$\Sigma = \mathcal{A}_d \Sigma \mathcal{A}_d^T + \Sigma_v - \mathcal{A}_d \Sigma \mathcal{C}_d^T (\mathcal{C}_d \Sigma \mathcal{C}_d^T + \Sigma_w)^{-1} \mathcal{C}_d \Sigma \mathcal{A}_d^T. \quad (12)$$

In summary, the full LQG solution is found in two steps: first solve a deterministic control problem establishing the optimal regulator in the CSI case, and then solve a MV estimation problem to find $\hat{x}_{k|k-1}$ in the ISI case (see Fig. 3 below).

To evaluate $\mathcal{J}^c(u)$ in Eq. (2), three actions are required; they are to be followed in the subsequent sections:

1. Further develop the quadratic criterion to retrieve the weighting matrices Q , R and S by comparing Eq. (2) to Eq. (6).
2. With these determine the state feedback and the observer gains $\mathcal{K}^{\text{opt}}$ and $\mathcal{L}^{\text{opt}}$, which are found from the solution of a control and an estimation Riccati equation, respectively, upon full construction of the state-space system of Eq. (5) (in Subsections 4.A and 4.B).
3. Finally, the residual phase variance can be analytically evaluated (in Subsection 4.C).

The next section establishes the LQG control solution for AO systems with infinitely fast DM as a first step toward the more complex case with slower DM.

3. LQG CONTROL FOR AO SYSTEMS WITH INFINITELY FAST MIRRORS

The LQG solution for the MV AO control problem that is extended here has been previously established in the infinitely fast DM case [5,6,15]. For introductory purposes, it is recalled below. The CSI case is considered first.

A. Solution in the CSI Case

When the DM's transient response is negligible compared with the sampling period T_s so that its dynamics can be reduced to its DC gain matrix $N \in \mathbb{R}^{n \times m}$ (the so-called DM influence matrix), then

$$\phi^{\text{cor}}(t) = Nu_k, \quad \forall kT_s \leq t < (k+1)T_s. \quad (13)$$

As shown in [6], the globally optimal MV control—that is, the discrete control minimizing the continuous-time criterion $J^c(u)$ —is obtained by minimizing the degenerate (because the criterion is independent of the phase temporal evolution model, just the inter-sample phase mean value is used) equivalent discrete-time MV criterion [5,15], denoted by J_{ifm}^d , with subscript standing for infinitely fast mirror,

$$J_{\text{ifm}}^d(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\bar{\phi}_{k+1}^{\text{tur}} - Nu_k\|^2, \quad (14)$$

where $J^c(u)$ relates to $J_{\text{ifm}}^d(u)$ by

$$J^c(u) = J_{\text{ifm}}^d(u) + \delta J(T_s), \quad (15)$$

with $\delta J(T_s)$ the insurmountable error due to the use of averaged variables instead of continuous ones [16]:

$$\delta J(T_s) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \left(\frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t) - \bar{\phi}_{k+1}^{\text{tur}}\|^2 dt \right). \quad (16)$$

For the commonly used frame rates of several hundred hertz, $\delta J(T_s)$ is small and normally accounted for in the total error budget.

For the sake of simplicity, it shall be assumed that the DM influence matrix N is full column rank, in other words that $N^T N$ is invertible. Since $J_{\text{ifm}}^d(u)$ relates to $J^c(u)$ by a term independent of the control decision u in Eq. (15), it follows that $\arg \min_{\mathcal{U}} J^c(u) = \arg \min_{\mathcal{U}} J_{\text{ifm}}^d(u)$, with $\mathcal{U}$ the set of admissible controls, which establishes the equivalence of both criteria.

The optimal control decisions u_k in the CSI case are hence given by

$$u_k = \Theta \bar{\phi}_{k+1} = (N^T N)^{-1} N^T \bar{\phi}_{k+1}^{\text{tur}}, \quad (17)$$

with $\Theta \in \mathbb{R}^{m \times n}$ the orthogonal projector of phase onto the DM's space.

Two interesting features of the CSI control are worth noting: (1) it does not depend on any particular assumption on the turbulent phase and (2) it effectively redefines “complete state information” as the advance knowledge not of the whole turbulent phase trajectory over the next sampling interval but rather of its average value $\bar{\phi}_{k+1}^{\text{tur}}$. The latter is specially interesting for the ISI case covered next.

B. Solution in the ISI Case

In the ISI case the state is estimated from WFS measurements. These measurements can also be expressed as functions of the average phase $\bar{\phi}^{\text{tur}}$ since light flux emanating from a guide star is commonly integrated during the detector exposure time. Assuming a linear response, their output can be modeled to a first approximation as

$$y_k = D \bar{\phi}_{k-1}^{\text{res}} + w'_k = D \bar{\phi}_{k-1}^{\text{tur}} - DNu_{k-2} + w'_k, \quad (18)$$

where w'_k is a zero-mean, Gaussian white noise $w'_k \sim \mathcal{N}(0, \Sigma_{w'})$. The subscript indexes used in Eq. (18) are thoroughly explained in Fig. 2. The assumptions on the WFS read-out and processing delays are not limiting; the approach presented here could easily be adapted to cope with a different temporal diagram. In Eq. (18), the linear operator $D \in \mathbb{R}^{s \times n}$ is a device-specific phase-to-WFS influence matrix and N is defined in Eq. (13).

As a consequence, *any* stochastic model of $\bar{\phi}^{\text{tur}}$ will enable one to construct an “exhaustive control-oriented” model for the AO loop, i.e., one from which the globally optimal MV control can be derived using standard LQG procedures.

Assuming that $\bar{\phi}^{\text{tur}}$ is a wide-sense-stationary random process, the separation principle applies; the MV control then becomes

$$u_k = \Theta \hat{\bar{\phi}}_{k+1|y_k}^{\text{tur}}, \quad (19)$$

where $\hat{\bar{\phi}}_{k+1|y_k}^{\text{tur}}$ denotes the conditional expectation of $\bar{\phi}_{k+1}$ with respect to the sequence of all measurements available at $t = kT_s$, namely, $\mathcal{Y}_k = \{y_0, \dots, y_k\}$. In order to recast this MV estimation problem into the standard LQG formulation in Section 2, define z as a pseudo-measurement that relates to y by

$$z_k = y_{k+1} = D \bar{\phi}_k^{\text{tur}} - DNu_{k-1} + w_k \quad (20)$$

with $w_k = w'_{k+1}$. This allows one to account for the delay resulting from measurement and control processing time in a way that is fully coherent with the formulation in Section 2. To achieve a mathematically rigorous treatment, in what follows, the notation $\hat{x}_{k|k-1}$ will denote the conditional expectation of x_k with respect to $\mathcal{Z}_{k-1} = \{z_0, \dots, z_{k-1}\} = \mathcal{Y}_k$.

Any state-space model meant to describe these pseudo-measurements should accommodate $\bar{\phi}_{k+1}^{\text{tur}}$ to compute the optimal control decisions in Eq. (17) and further $(\bar{\phi}_k^{\text{tur}}, u_{k-1})$ in Eq. (20). Consider the average phase temporal evolution is given by a first-order auto-regressive model:

$$\bar{\phi}_{k+1}^{\text{tur}} = \mathcal{A}_{\text{tur}} \bar{\phi}_k^{\text{tur}} + v_k. \quad (21)$$

Though these simplistic models cannot thoroughly represent Taylor's frozen flow hypothesis, they are accurate enough to grasp the spatial (inter-modal) statistics and the short-term correlations. With slight variations that relate to the basis considered, all the cited papers in Section 1 use auto-regressive (AR) models. These models prove quite powerful even in laboratory experiments using rotating phase screens (quasi-pure frozen flow) [17].

Gathering all in a complete state, the full discrete-time state-space becomes

$$\begin{pmatrix} \bar{\phi}_{k+2}^{\text{tur}} \\ \bar{\phi}_{k+1}^{\text{tur}} \\ u_k \end{pmatrix} = \begin{pmatrix} \mathcal{A}_{\text{tur}} & 0 & 0 \\ I & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \bar{\phi}_{k+1}^{\text{tur}} \\ \bar{\phi}_k^{\text{tur}} \\ u_{k-1} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ I \end{pmatrix} u_k + \begin{pmatrix} I \\ 0 \\ 0 \end{pmatrix} v_k, \quad (22)$$

with

$$z_k = D(0 \quad I \quad -N)x_k + w_k. \quad (23)$$

Using z_k instead of y_k results in a more compact state model than that of [5]. The control decisions are found from

$$u_k = \Theta(I \quad 0 \quad 0)\hat{x}_{k|k-1}, \quad (24)$$

where the predictor version of the controller uses information up to z_{k-1} or equivalently up to y_k . This ensures that the loop delays are correctly taken into account in the complete model of Eqs. (22)–(24).

Expanding Eq. (14) and comparing it with Eq. (6), the LQ weighting matrices are readily

$$Q = H^T H \geq 0, \quad S = -NH^T, \quad R = N^T N > 0, \quad (25)$$

and $H = (I \quad 0 \quad 0)$. This yields an equivalent discrete-time criterion of the form

$$J_{\text{ifm}}^d(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \begin{pmatrix} \bar{\phi}_{k+1}^{\text{tur}} \\ u_k \end{pmatrix}^T \begin{pmatrix} I & -N \\ -N^T & N^T N \end{pmatrix} \begin{pmatrix} \bar{\phi}_{k+1}^{\text{tur}} \\ u_k \end{pmatrix}. \quad (26)$$

In this case, the optimal control gain, as shown above, can be computed directly as an orthogonal projection. The optimal observer gain (Kalman gain), on the other hand, is computed from an estimation Riccati equation.

For the complete description of the ISI case, see [5,6,15]. For the treatment of the general AR model of higher order refer to [5].

The compactness and computational burden of the model of Eqs. (22)–(24) has been addressed in [6,16]. It has later been successfully tested experimentally, together with an extension to vibration rejection [17].

In next section, the results so far obtained will be generalized to a LTI DM model with non-negligible dynamics using the same methodology.

4. LQG CONTROL FOR AO SYSTEMS WITH NON-NEGLIGIBLE DM DYNAMICS

In the general case, when the DM transient response is of duration greater than zero, the derivation of the optimal regulator must include knowledge about the temporal evolution of the DM deformation. The equivalent discrete-time criterion naturally involves additional turbulent-phase variables and the DM's internal state(s). The optimal MV control then becomes a non-degenerate LQ problem, the solution of which can no longer be constructed independently from the continuous-turbulent-phase model.

Hypothesis 1: Let the DM correction phase be given by

$$\phi^{\text{cor}} = Np(t), \quad (27)$$

with $p(t)$ the DM instantaneous deformation on a basis of normalized DM influence functions. It has been assumed that the DM shapes are all within $\text{Im}(N)$, the space spanned by the static DM influence functions. A larger discussion on the construction of this modes is provided in Appendix C.

The DM instantaneous deformation is the output of a continuous-time LTI state-space model in the form

$$\begin{cases} \dot{x}^m(t) = \mathcal{A}_m x^m(t) + \mathcal{B}_m u(t) \\ p(t) = \mathcal{C}_m x^m(t) + \mathcal{D}_m u(t) \end{cases}, \quad (28)$$

with the following assumptions: (a) $\mathcal{A}_m$ is a stability matrix (all eigenvalues have strict negative real part); (b) the pair $(\mathcal{A}_m, \mathcal{C}_m)$ is observable, and (c) the DC gain between u and p is unitary. Note that assumptions (a)–(c) should essentially be regarded as the requirement that the DM model has been constructed according to the control theory's good practices.

Going back to Eq. (1) and developing the square with use of Eq. (27) yields for the term

$$\begin{aligned} \|\phi^{\text{tur}}(t) - \phi^{\text{cor}}(t)\|^2 &= \phi^{\text{tur}}(t)^T \phi^{\text{tur}}(t) \\ &\quad + p(t)^T N^T N p(t) - 2\phi^{\text{tur}}(t)^T N p(t). \end{aligned} \quad (29)$$

For a constant input $u(t) = u_k$, the exact solution of $p(t)$ given in Appendix A, $\forall t \in [kT_s, (k+1)T_s]$ is

$$p(t) = \mathcal{C}_m e^{(t-kT_s)\mathcal{A}_m} x_k^m + (I + \mathcal{C}_m e^{(t-kT_s)\mathcal{A}_m} \mathcal{A}_m^{-1} \mathcal{B}_m) u_k. \quad (30)$$

Replacing $p(t)$ in Eq. (29) by its expression in Eq. (30) and rearranging terms, the continuous-time criterion $J^c(u)$ is obtained from the sum of two contributions of which only one is a function of the control decisions u :

$$J^c(u) = J_{\text{tur}} + J_{\text{dyn}}^d(u). \quad (31)$$

By adding/subtracting terms independent of u it is guaranteed that

$$\arg \min_u J^c(u) = \arg \min_u J_{\text{dyn}}^d(u). \quad (32)$$

Proceeding with the integration of Eq. (29) with $p(t)$ defined in Eq. (30), it is found that three components are required to determine $J_{\text{dyn}}^d(u)$. Concatenate them and define the performance state $z_k^{\text{crit}} = \mathcal{C}_d^{\text{crit}} x_k$,

$$z_k^{\text{crit}} \triangleq \begin{pmatrix} \bar{\phi}_{k+1}^{\text{tur}} \\ \bar{\varphi}_{k+1}^{\text{tur}} \\ x_k^m \end{pmatrix} \triangleq \begin{pmatrix} \frac{1}{T_s} \int_0^{T_s} \phi^{\text{tur}}(kT_s + s) ds \\ \frac{1}{T_s} \int_0^{T_s} e^{s\mathcal{A}_m^T} \mathcal{C}_m^T N^T \phi^{\text{tur}}(kT_s + s) ds \\ x^m(kT_s) \end{pmatrix}, \quad (33)$$

such that

$$J_{\text{dyn}}^d(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^M \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}^T \begin{pmatrix} \bar{Q}_1 & \bar{S} \\ \bar{S}^T & \bar{R} \end{pmatrix} \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}. \quad (34)$$

A note on the composition of z_k^{crit} . Compared with Eq. (14), where only $\bar{\phi}_{k+1}^{\text{tur}}$ intervenes, it now has the following components: the average phase $\bar{\phi}_{k+1}^{\text{tur}}$, the weighted temporal average $\bar{\varphi}_{k+1}^{\text{tur}}$ (weighted by the DM impulse response) and the DM internal state x_k^m . These components appear as natural, since they pop up directly from the development of the quadratic summation: a quadratic term on the phase, a quadratic term on the DM, and a cross term.

Further details on the derivation of Eqs. (31)–(33) are found in Appendix A. Terms independent of u were explicitly added to ensure positive definiteness of the discrete-time criterion J_{dyn}^d , which can therefore be put in the form and conditions of Eq. (6). Since in so doing Eq. (32) stands, the optimal controls u can likewise be found by minimizing J_{dyn}^d in Eq. (34) or, equivalently, $J^c(u)$ in Eq. (31). The LQ weighting matrices are

$$\bar{Q}_1 \triangleq \begin{pmatrix} 2 & & \\ -NN^T & 0 & 0 \\ \varepsilon & & \end{pmatrix} \geq 0, \quad (35)$$

$$\bar{S} \triangleq \begin{pmatrix} -N \\ -\mathcal{A}_m \mathcal{B}_m \\ S_0 \end{pmatrix}, \quad \bar{R} = R_0, \quad (36)$$

where

$$R_0 \triangleq \frac{1}{T_s} \int_0^{T_s} (I + C_m e^{s\mathcal{A}_m} \mathcal{A}_m^{-1} \mathcal{B}_m)^T N^T N (I + C_m e^{s\mathcal{A}_m} \mathcal{A}_m^{-1} \mathcal{B}_m) ds, \quad (37)$$

$$S_0 \triangleq \frac{1}{T_s} \int_0^{T_s} e^{s\mathcal{A}_m^T} C_m^T N^T N (I + C_m e^{s\mathcal{A}_m} \mathcal{A}_m^{-1} \mathcal{B}_m) ds, \quad (38)$$

$$Q_0 \triangleq \frac{1}{T_s} \int_0^{T_s} e^{s\mathcal{A}_m^T} C_m^T N^T N C_m e^{s\mathcal{A}_m} ds. \quad (39)$$

Assuming that $z_k^{\text{crit}} = C_d^{\text{crit}} x_k$ is a function of the state, matrices $\bar{Q}$, $\bar{S}$, and $\bar{R}$ relate to Q , S , and R in Eq. (6) by taking $Q = (C^{\text{crit}})^T \bar{Q} C^{\text{crit}}$, $S = (C^{\text{crit}})^T \bar{S}$, and $R = \bar{R}$. C_d^{crit} will be defined in Subsection 4.B.

In Eq. (35), $\varepsilon > 0$ is a small constant affecting terms independent of u to guarantee positive definiteness of the form in Eq. (7). Demonstration and derivation of Eqs. (33)–(39) are provided in Appendix A.

Following the construction of a model to generate the states in Eq. (33), the performance criterion of Eq. (31) is evaluated using Eq. (7). In addition, the optimal regulator is also derived from the solution to the LQ problem. These points are covered next.

A. Control-Oriented Model for the CSI Case

A standard result in control theory is that a LQ problem can be solved constructively when z^{crit} in Eq. (33) can be obtained from the output of a discrete-time linear stochastic model.

With one such model and the so-far-derived performance state of Eq. (33) and the LQ weighting matrices of Eqs. (35)–(39), the equivalent LQ criterion $J_{\text{dyn}}^d(u)$ in Eq. (34) can be evaluated.

Before continuing, a further assumption on the disturbance is considered.

Hypothesis 2. The disturbance can be modeled as the output of a stochastic, linear and time-invariant (LTI)

system whose input is a Gaussian white noise with power spectral distribution (PSD) Σ_v ,

$$\begin{cases} \dot{x}^{\text{tur}}(t) = \mathcal{A}_{\text{tur}} x^{\text{tur}}(t) + v(t) \\ \phi_{\text{tur}}(t) = C_{\text{tur}} x^{\text{tur}}(t) \end{cases}. \quad (40)$$

This step in the analysis relies on the certainty equivalence principle. It is based on the hypothesis that the LQ solution to the deterministic system, obtained by replacing stochastic variables by their expected values is also the optimal solution to the original stochastic system [18].

The optimal LQ regulator is of the form

$$u_k = -\mathcal{K}^{\text{opt}} x_k, \quad (41)$$

with $\mathcal{K}^{\text{opt}}$ defined as

$$\mathcal{K}^{\text{opt}} = (R + \mathcal{B}_d^T P_{\infty} \mathcal{B}_d)^{-1} (\mathcal{B}_d^T P_{\infty} \mathcal{A}_d + S), \quad (42)$$

determined from the (unique) solution to the algebraic discrete-time control Riccati equation (ARE):

$$\begin{aligned} P_{\infty} &= Q_1 + \mathcal{A}_d^T P_{\infty} \mathcal{A}_d - (\mathcal{A}_d^T P_{\infty} \mathcal{A}_d + S) \\ &\quad \times (R + \mathcal{B}_d^T P_{\infty} \mathcal{B}_d)^{-1} (\mathcal{B}_d^T P_{\infty} \mathcal{A}_d + S). \end{aligned} \quad (43)$$

In the remainder of this paper, a sub-optimal regulator that neglects the mirror's dynamics is compared with the optimal regulator $\mathcal{K}^{\text{opt}}$. In Section 3 this sub-optimal solution in this new context has been found optimal should the mirror be infinitely fast; in that case a simple orthogonal projection onto the mirror space was performed. Moreover, the presence of DM dynamics is not accounted for during the estimation step.

B. Control-Oriented Model for the ISI Case

In the ISI case, the conditional expectation of x_k , $\hat{x}_{k|k-1}$, is determined by a recursive Kalman filter [19], as explained in Section 2.

Focusing on the specific arrangement of measurements and controls in AO (Fig. 2) the measurement at $t = kT_s$ is given by

$$y_k = D \frac{1}{T_s} \int_{(k-2)T_s}^{(k-1)T_s} (\phi^{\text{tur}}(t) - Np(t)) dt + w'_k \quad (44)$$

$$\begin{aligned} &= D \bar{\phi}_{k-1}^{\text{tur}} - DN \left[\frac{C_m}{T_s} (e^{T_s \mathcal{A}_m} - I) \mathcal{A}_m^{-1} x_{k-2}^m \right. \\ &\quad \left. + \left(\frac{C_m}{T_s} (e^{T_s \mathcal{A}_m} - I) \mathcal{A}_m^{-2} \mathcal{B}_m + I \right) u_{k-2} \right] + w'_k. \end{aligned} \quad (45)$$

This is done by replacing $p(t)$ by its expression in Eq. (30) and further simplifying.

The complete state-space model of Eq. (5) with extended state is now

$$x_k \triangleq \begin{pmatrix} x_{k+1}^{\text{tur}} \\ \bar{\phi}_{k+1}^{\text{tur}} \\ \bar{\phi}_{k+1}^{\text{tur}} \\ x_k^m \\ \bar{\phi}_k^{\text{tur}} \\ x_{k-1}^m \\ u_{k-1} \end{pmatrix} \quad \mathcal{A}_d \triangleq \begin{pmatrix} e^{T_s A_{\text{tur}}} & 0 & 0 & 0 & 0 & 0 & 0 \\ \Xi & 0 & 0 & 0 & 0 & 0 & 0 \\ \Phi & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{T_s A_m} & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (46)$$

$$\mathcal{B}_d \triangleq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \mathcal{A}_m^{-1}(e^{T_s A_m} - I)\mathcal{B}_m \\ 0 \\ 0 \\ I \end{pmatrix} \quad \Gamma \triangleq \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (47)$$

$$\mathcal{C}_d \triangleq \begin{pmatrix} 0 & 0 & 0 & 0 & D & -\frac{DNC_m}{T_s}(e^{T_s A_m} - I)\mathcal{A}_m^{-1} & \dots \\ -DN\left(\frac{C_m}{T_s}(e^{T_s A_m} - I)\mathcal{A}_m^{-2}\mathcal{B}_m + I\right) \end{pmatrix}, \quad \mathcal{D}_d = 0, \quad (48)$$

$$\mathcal{C}_d^{\text{crit}} \triangleq \begin{pmatrix} 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 \end{pmatrix}. \quad (49)$$

Matrices Ξ and Φ are derived in Appendix B.

Analogously to what has been done in Section 3, to re-introduce the overlooked delay, one uses all information available until $t=(k-1)T_s$ instead of all the information available at $t=kT_s$. As a consequence the Kalman filter is used in predictor form, i.e.,

$$u_k = -\mathcal{K}^{\text{opt}} \hat{x}_{k|k-1}. \quad (50)$$

It is equivalent to associate the delay with the controls instead of the measurements. The two-step delay is thus kept correctly. A version that employs directly y_k instead of the pseudo-measurement z_k can be found in [10] at the expense of a larger state-space system. State-space compactness is, for obvious reasons, more attractive for real-time implementation.

A full state-space representation is now available for both the infinitely fast and the non-negligible DM dynamics cases. With the weighting matrices Q , R , and S the optimal regulator and Kalman gains can now be determined. These define the full structure of the LQ-LQG solution.

The evaluation of $J^c(u)$ in Eq. (1) is still missing. It is detailed in Subsection 4.C for the general linear state-feedback regulator $\mathcal{K}$.

C. Analytical Performance Evaluation

Since ϕ^{tur} is an ergodic process, the average disturbance energy over an infinite horizon is almost surely equal to its expected value [20], i.e.,

$$\lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{\text{tur}}(t)\|^2 dt = \mathbb{E}(\|\phi^{\text{tur}}(t)\|^2), \quad (51)$$

at any time t , including sampling times $t=kT_s$, $k \in \mathbb{N}$. $\mathbb{E}(\cdot)$ denotes mathematical expectation. Thus,

$$\lim_{\tau \rightarrow +\infty} \frac{1}{\tau} \int_0^\tau \|\phi^{\text{tur}}(t)\|^2 dt = \text{trace}(\mathcal{C}'_{\text{tur}} \mathcal{C}'_{\text{tur}} \Sigma_x), \quad (52)$$

where $\mathcal{C}'_{\text{tur}} \triangleq (\mathcal{C}_{\text{tur}} \quad 0 \quad \dots \quad 0)$.

The criterion of Eq. (31) can be looked at as the expectation of a quadratic form. Replacing z_k^{crit} by $\mathcal{C}_d^{\text{crit}} x_k$, one gets

$$\begin{aligned} J^c(u) &\stackrel{a.s.}{=} \mathbb{E}(\|\phi^{\text{tur}}(t)\|^2) + \mathbb{E}((z_k^{\text{crit}})^T \bar{Q} z_k^{\text{crit}} + u_k^T R u_k - 2(z_k^{\text{crit}})^T \bar{S} u_k) \\ &\stackrel{a.s.}{=} \mathbb{E}(\|\phi^{\text{tur}}(t)\|^2) + \mathbb{E}(x_k^T Q x_k + \hat{x}_{k|k-1}^T R \mathcal{K}^T R \mathcal{K} \hat{x}_{k|k-1} \\ &\quad + 2x_k^T S \mathcal{K} \hat{x}_{k|k-1}). \end{aligned} \quad (53)$$

Property 1: Consider the system

$$\zeta_{k+1} = A \zeta_k + \eta_k \quad (54)$$

with A_f stable, i.e., with all its eigenvalues $|\lambda_i| \leq 1$. Then ζ_k will converge to a Gaussian process whose variance is the solution to the discrete Lyapunov equation [18,19]

$$\Sigma_\zeta = A \Sigma_\zeta A^T + \Sigma_\eta, \quad (55)$$

where Σ_ζ is the steady-state covariance matrix of the zero-mean stochastic state ζ . With Σ_ζ and $P \geq 0$ [18],

$$\mathbb{E}(\zeta^T P \zeta) = \text{trace}(P \Sigma_\zeta). \quad (56)$$

To evaluate the criterion in Eq. (53), define the augmented state $\zeta = (x_k^T, \hat{x}_{k|k-1}^T)^T$. With the control decisions $u_k = -\mathcal{K} \hat{x}_{k|k-1}$, the state recurrence equation is written

$$x_{k+1} = \mathcal{A}_d x_k + \mathcal{B}_d u_k + \Gamma v_k = \mathcal{A}_d x_k - \mathcal{B}_d \mathcal{K} \hat{x}_{k|k-1} + \Gamma v_k. \quad (57)$$

For the estimated state, using the Kalman filter equations [19] one gets

$$\begin{aligned} \hat{x}_{k+1|k} &= \mathcal{A}_d \hat{x}_{k|k-1} + \mathcal{B}_d u_k + \mathcal{L}(z_k - \mathcal{C}_d \hat{x}_{k|k-1}) \\ &= \mathcal{A}_d \hat{x}_{k|k-1} + \mathcal{B}_d u_k + \mathcal{L}(\mathcal{C}_d x_k - \mathcal{C}_d \hat{x}_{k|k-1} + w_k) \\ &= (\mathcal{A}_d - \mathcal{B}_d \mathcal{K} - \mathcal{L} \mathcal{C}_d) \hat{x}_{k|k-1} + \mathcal{L} \mathcal{C}_d x_k + \mathcal{L} w_k. \end{aligned} \quad (58)$$

Rearranging terms, from Eq. (57) and Eq. (58) the augmented state-space system is therefore

$$\begin{pmatrix} x_{k+1} \\ \hat{x}_{k+1|k} \end{pmatrix} = A \begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix} + \begin{pmatrix} \Gamma & 0 \\ 0 & \mathcal{L} \end{pmatrix} \begin{pmatrix} v_k \\ w_k \end{pmatrix}, \quad (59)$$

where

$$A \triangleq \begin{pmatrix} \mathcal{A}_d & -\mathcal{B}_d \mathcal{K} \\ \mathcal{L} \mathcal{C}_d & \mathcal{A}_d - \mathcal{B}_d \mathcal{K} - \mathcal{L} \mathcal{C}_d \end{pmatrix}. \quad (60)$$

From property 1, the covariance matrix Σ_ζ of the augmented state ζ of the form

$$\Sigma_\zeta = \begin{pmatrix} \Sigma_x & \Sigma_{x_k x_{k|k-1}} \\ \Sigma_{x_k x_{k|k-1}} & \Sigma_{x_{k|k-1}} \end{pmatrix} \quad (61)$$

is determined by solving the associated discrete-time Lyapunov equation $\Sigma_\zeta = A_\zeta \Sigma_\zeta A_\zeta^\top + \Sigma_\eta$. The state noise covariance matrix,

$$\Sigma_\eta \triangleq \begin{pmatrix} \Gamma \Sigma_v \Gamma^\top & 0 \\ 0 & \mathcal{L} \Sigma_w \mathcal{L}^\top \end{pmatrix}, \quad (62)$$

is found from Eqs. (57) and (58) using the fact that v_k and w_k are uncorrelated with covariance matrices Σ_v and Σ_w , respectively.

With use of the second result in property 1, the continuous-time criterion $J^c(u)$ can be written as

$$J^c(u) \stackrel{a.s.}{=} E(\zeta^\top W \zeta) \stackrel{a.s.}{=} E \left(\begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix}^\top \begin{pmatrix} C_{\text{tur}}' C_{\text{tur}}^\top + Q & SK \\ K^\top S^\top & K^\top R K \end{pmatrix} \begin{pmatrix} x_k \\ \hat{x}_{k|k-1} \end{pmatrix} \right) \stackrel{a.s.}{=} \text{trace}(W \Sigma_\zeta). \quad (63)$$

Note the use of Q instead of Q_1 in Eq. (63), which directly adds the second and third terms in Eq. (31). Refer to Appendix A for details on this operation.

To summarize, the weighting matrices Q , R , and S and both the regulator $\mathcal{K}^{\text{opt}}$ and the observer gain $\mathcal{L}^{\text{opt}}$ were determined. In the optimal case they require the resolution of the associated Riccati equations. With Eq. (63) the performance criterion $J_c(u)$ can be quantified for both the optimal and the sub-optimal cases using an analytical approach. This tool provides a means to synthesize the optimal regulator for AO systems with non-negligible dynamics and to study the impact of system parameters on the overall performance. This will hold particular interest for system designing and prototyping.

5. ELT TIP/TILT CONTROL

The examples in this section concentrate on the tip/tilt DM (TTDM) modes, though the state-space model of Section 4 is general and extensible to any number of modes; on the one hand, this is a simple and representative case; and on the other, TTs are very energetic optical modes [21]. To this is added the fact that many AOs use a two-stage correction composed of a TT mount and a DM; thus TT modes can be regarded as a somewhat special set.

More recently, the migration from active to adaptive telescopes with the inclusion of a DM in the optical path (see for example [22]) and the designs foreseen for the ELTs make an appeal for ever more accurate regulators. These are needed to drive structurally massive AOSs affected by large disturbances induced by the telescope's mechanical vibrations. Due to structure stiffness, low-order optical modes tend to be particularly affected by wind buffeting, thus emphasizing the importance of accurate TT control [23]. To this is added the fact that down-

scaled finite-element models show that the first flexible eigenmodes do present a considerable TT component. Computational requirements for the TT small set remain sustainable despite the extended state space built for the ISI case. With no loss of generality, $p(t)$ is considered here as the equivalent on-sky angle of the TTDM with the flat reference position.

Having thoroughly defined the weighting matrices Q , R , and S in Eq. (7), the gains $\mathcal{K}^{\text{opt}}$ and $\mathcal{L}^{\text{opt}}$, and the control-oriented state space of Eqs. (46)–(48), the performance of the system can be evaluated for the optimal regulator in Eq. (50) and the sub-optimal regulator of Eq. (17) that neglects DM dynamics. Two different cross-checking tools are used:

- Analytical: the performance is evaluated using the analytical results of Subsection 4.C. Matrices A_d^{sys} , B_d^{sys} , and C_d^{sys} in Fig. 3 (which include both the disturbance and the DM model) are defined beforehand. The control-oriented model based on these is said to be coherent with the disturbance and DM models. Additionally, the control-oriented state space can be converted into transfer functions. The residual variance is found from the integration over frequencies. Such transfer functions provide a valuable graphical interpretation of the results complementary to the analytical code.

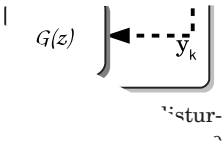
- Monte Carlo end-to-end simulation: it uses continuous temporal models that translate the block diagram of Fig. 3. The fine discretization of such models is beyond the scope of the discussion here. Variances are then determined over sufficiently long realizations to guarantee statistical convergence. The Monte Carlo simulator input disturbances are time series of the modal coefficients of $\phi^{\text{tur}}(t)$, in two configurations depicted in Fig. 4: (1) control-oriented model incoherent with disturbance trajectories, where the wind-buffeting time series, dubbed WB series, are determined, for example, from finite-element analysis and (2) control-oriented model coherent with the identified disturbance model, where the time series are the output of an n th-order model whose parameters are defined based on the WB time series. Such a model is encapsulated in the identified disturbance model (IDM) block in Fig. 4 and is used to synthesize a coherent control-oriented model.

Subsection 5.D.1 deals with the coherent case comparing the optimal to the sub-optimal regulator. Additionally it validates the Monte Carlo simulator results. It is later used in Subsection 5.D.2 with model-independent disturbance trajectories. In so doing, modeling errors related to the disturbance are added to the total error budget unlike in the previous cases where only temporal and estimation errors were quantified. Moreover, the identification procedure can be validated should the modeling errors vanish or be small compared with their estimation and temporal counterparts.

A. Disturbance Linear Model

Motivated by the one-step correlation function [see in Appendix B Eq. (B12)], the choice of the continuous-time disturbance model is based on fitting the short-term temporal correlation function at time scales close the sampling time T_s .

VB series



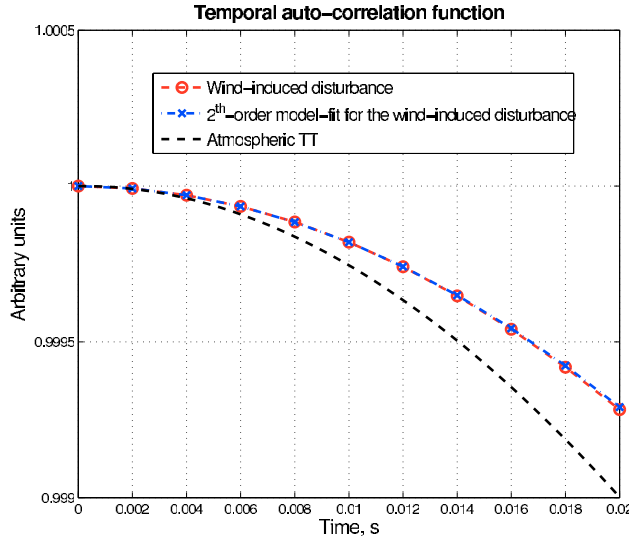


Fig. 6. (Color online) Wind-buffeting auto-correlation curve and that of a second-order model that fits it. The auto-correlation of the atmospheric TT is shown for comparison.

frequency behavior and cutoff frequency as shown in Fig. 5. The short-term auto-correlation shown in Fig. 6 points out the possibility of closely modeling the disturbance behavior by a second-order model (due to the negative curvature in the vicinity of zero; a first-order model with an exponential auto-correlation function would be unadapted). Although differences are found in the long run, short-term auto-correlation is preserved.

In the following, only WB disturbances are considered due to the high energy involved compared with the atmospheric disturbances.

B. DM Linear Model

Second-order DM models are considered. They closely represent the dynamical behavior of a broad range of physical devices. Simpler models of the first order were addressed in [10].

It is assumed the DM model follows the second-order differential equation

$$\ddot{p}(t) + 2\xi\omega_n\dot{p}(t) + \omega_n^2 p(t) = \omega_n^2 u(t), \quad (64)$$

where ξ is the damping coefficient and $\omega_n = 2\pi f_n$ is the natural resonance pulsation (in radians per second). Mass-spring-damper systems are also described by this differential equation, with the parameters ξ and ω_n determined from the mass, the spring constant, and the viscous coefficient.

In terms of state spaces, Eq. (64) converts to

$$\mathbf{A}_m = \begin{pmatrix} 0 & 1 \\ -\omega_n^2 & -2\xi\omega_n \end{pmatrix}, \quad \mathbf{B}_m = \begin{pmatrix} 0 \\ \omega_n^2 \end{pmatrix},$$

$$\mathbf{C}_m = (1 \quad 0), \quad \mathbf{D}_m = 0. \quad (65)$$

According to ξ a resonant frequency may be observed at $f_n = 1/2\omega_n\pi\sqrt{1-\xi^2}$. The corresponding frequency responses are depicted in Fig. 7 for three distinct values of f_n .

In addition, the problem being scalar, the state-space models can be converted into transfer functions, a complementary interpretation tool commonly used in AO.

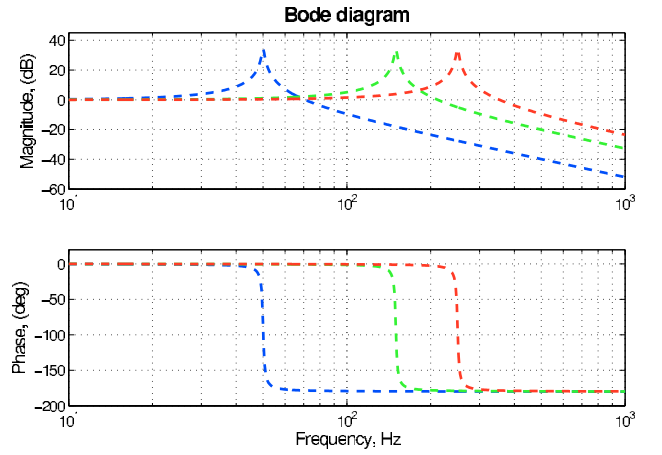


Fig. 7. (Color online) Bode diagram of the tip/tilt obtained from the finite-element mechanical model. Frequency abscissa numbers are in hertz for the DM with natural frequencies located at $f_n = \{50, 150, 250\}$ Hz and damping coefficient $\xi = 0.01$.

For illustration purposes, reference values of damping $\xi = 0.01$ are considered. At the resonant frequency $f_r = f_n\sqrt{1-\xi^2}$, amplitude response is amplified by a factor of 50. This naturally configures a very weakly damped system that tends to oscillate at f_r with slow energy-dissipation rate. Finite-element models built so far do not consider any supports that may in practice change these parameters. They are used here only for illustration purposes. A broader discussion on large DM models can be found in [27].

C. Frequency Simulations

Define the $\mathcal{Z}$ transform of turbulent phase $\Phi^{\text{tur}}(z)$, residual phase $\Phi^{\text{res}}(z)$, and noise $W(z)$, where z is a complex variable.

Both the phase-rejection transfer function (PRF) $\Phi^{\text{res}}(z)/\Phi^{\text{tur}}(z)$ and the noise-rejection transfer function (NRF) $\Phi^{\text{res}}(z)/W(z)$ can be determined by identification on the block structure of Fig. 1 and control loop of Fig. 3. The frequency responses are depicted in Figs. 8 and 9. Al-

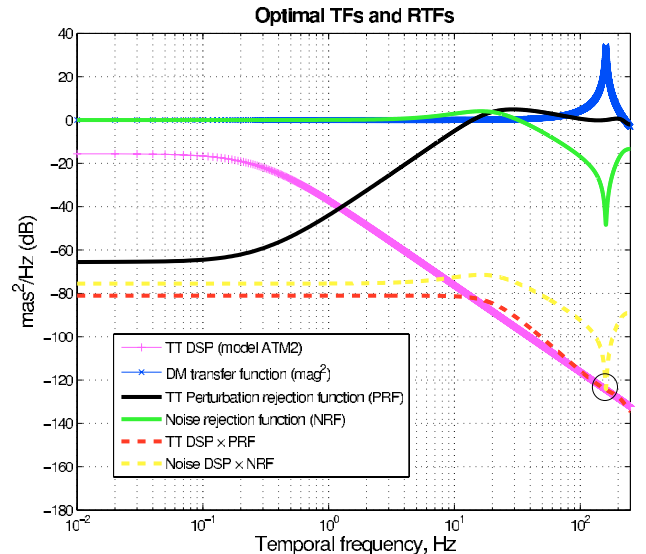


Fig. 8. (Color online) Optimal regulator's rejection transfer functions with the disturbance and the DM models.

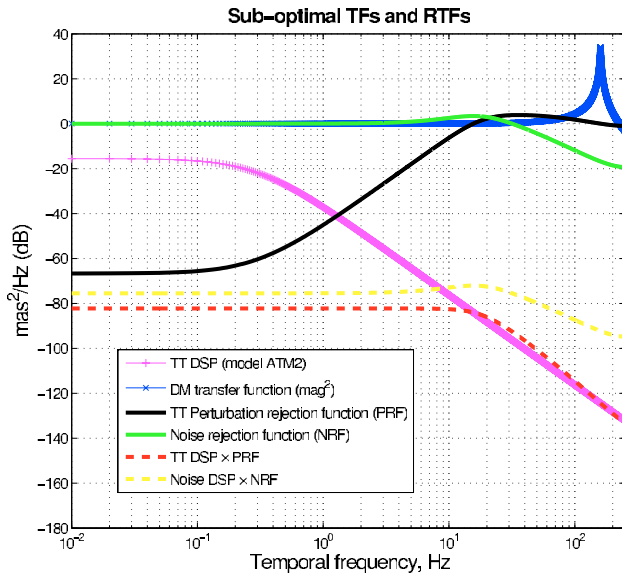


Fig. 9. (Color online) Suboptimal regulator's rejection transfer functions with the disturbance and the DM models. $T_s = 500$ Hz.

though the PRFs are very alike whatever regulator considered, the NRFs are quite different: in the optimal case the NRF damps noise at frequencies where it would tend to be largely amplified by the DM resonant frequency (the bell shape in the upper-right corners of Fig. 8 and Fig. 9). Note also that the noise rejection level at the DM resonant frequency f_r equals that for the disturbance itself. The two curves touch, with the shape of the NRF being the opposite of the bell-shaped DM frequency response. Moreover, the optimal PRF goes slightly below the 0 dB threshold in the band of frequencies where the DM TF goes above; on the sub-optimal side (recall it is the simple orthogonal projection of the predicted phase onto the DM space, described in Section 3, where DM dynamics are neglected), noise rejection is far less effective, as shown in Fig. 9; in all, the optimal regulator optimally rejects frequency by frequency the propagation of noise and disturbance through the AO loop.

D. Sample Numerical Results

1. Control-Oriented Model Coherent with Identified Disturbance Model

On a first attempt, consider that a model of the disturbance is available. In other words, consider that the disturbance $\phi^{\text{tur}}(t)$ in Fig. 4 is the output of the IDM block, whose inner model given by Eq. (40) is identified from the PSD of the wind-induced disturbance shown in Fig. 5, according to Subsection 5.A. The regulator is synthesized using the same model parameters, so it is coherent with the disturbance model. In this configuration, the modeling errors are not accounted for, only the temporal and estimation errors (the state x is estimated by a Kalman filter). It is therefore useful to investigate the performance attainable by the different regulators for several combinations of natural frequencies f_n , damping coefficients ξ , and noise levels. Throughout the simulations $T_s = 2$ ms has been adopted.

It was observed that for variations of damping coefficients around the reference $\xi = 0.01$, performance is very

close. For values $\xi > 0.1$ some gain is brought, but such weakly oscillating configurations are not considered here. Results are shown in Fig. 10 for a damping of $\xi = 0.01$ and a reference noise level of $\delta = 200$ nm rms at the edges of telescope's pupil.

Several conclusions are in order: using the optimal controller ensures the same performance $\forall f_n$, since DM dynamics is always correctly inverted. A plateau around 3.5 mas rms was found. On the sub-optimal controller's side, the residual phase is always greater. It varies between 4 and 6.5 mas rms. Moreover, the stability region is limited to a smaller set of natural resonant frequencies f_n in the range 100–220 Hz. Recall that the sub-optimal controller is to be used under the assumption that it stabilizes the loop. In approaching the stability limits, differences are to be found between the analytic and Monte Carlo simulations. Such differences are certainly due to poor statistical convergence in these extreme cases. Note that in the optimal case there is a very good match between the analytical and Monte Carlo simulations.

The differences in performance between the optimal and sub-optimal regulators have two sources: (1) on the observer side, ignoring the DM dynamics results in degraded state estimation, and (2) on the control side, the control decisions ignore the oscillating behavior of the DM surface. For example, taking into account the dynamics on the observer side only gives results in between the curves presented in Fig. 10.

One can get some insight into the actual difference of operation between regulators by observing their temporal behaviors. The sub-optimal controller, neglecting dynamics, will allow for noise propagation that will excite the DM at its resonant frequency. In such a case, the outcome is a “ringing” effect on the mirror trajectory centered at f_r , as depicted in Fig. 11. This effect is likely to lead to strong

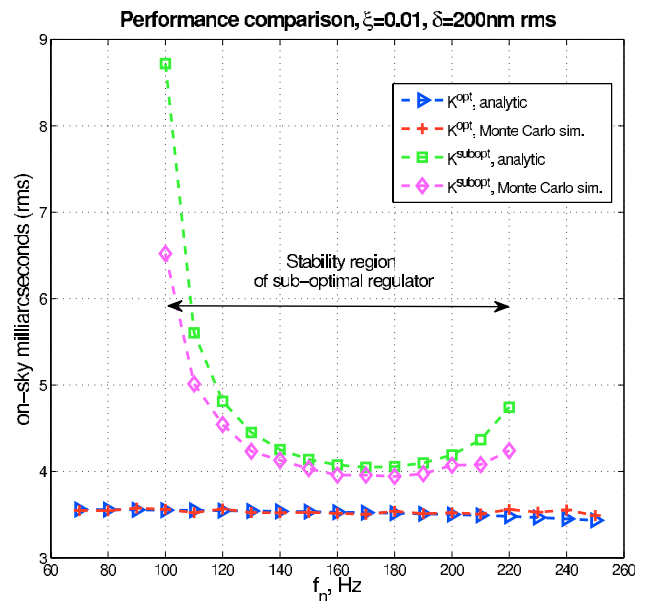


Fig. 10. (Color online) Performance comparison of the optimal and sub-optimal regulators. Analysis versus Monte Carlo simulation. The double arrow indicates the stability region of the sub-optimal solution. Outside these bounds the sub-optimal regulator cannot operate. $T_s = 500$ Hz, $\delta = 200$ nm at the edges of the telescope.

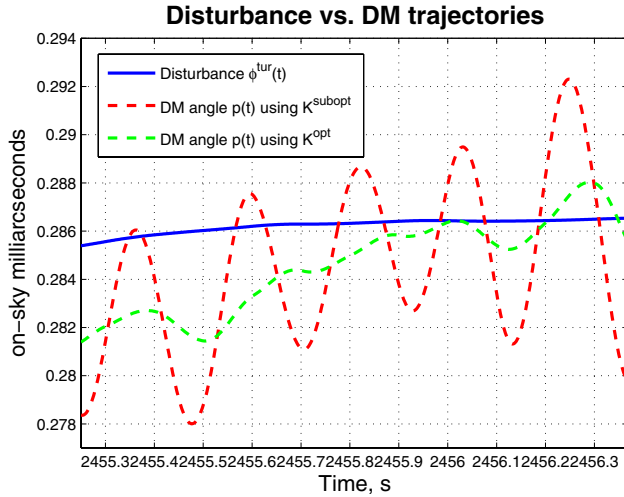


Fig. 11. (Color online) DM ringing effect. The difference in managing the strong oscillations likely to appear in weakly damped DM can be taken into account by the optimal regulator (light dashed curve). Sub-optimal solutions tend to present oscillatory responses of large amplitude (dark dashed curve). $T_s=500$ Hz and $f_n=160$ Hz.

oscillations and to grave losses in performance. In contrast, the optimal controller has a better noise rejection at the DM resonant frequency, thus avoiding large amplitude oscillations. This yields a temporal response that follows the disturbance more closely.

2. Control-Oriented Model Incoherent with Disturbance Model

Suppose now that the disturbance is a time-series WB of which the stochastic model is unknown. The disturbance parameters used in the control-oriented model, though identified from the disturbance PSD in Fig. 5 as explained in Subsection 5.A, are structurally incoherent with the input. Such full-picture configuration allows one to add

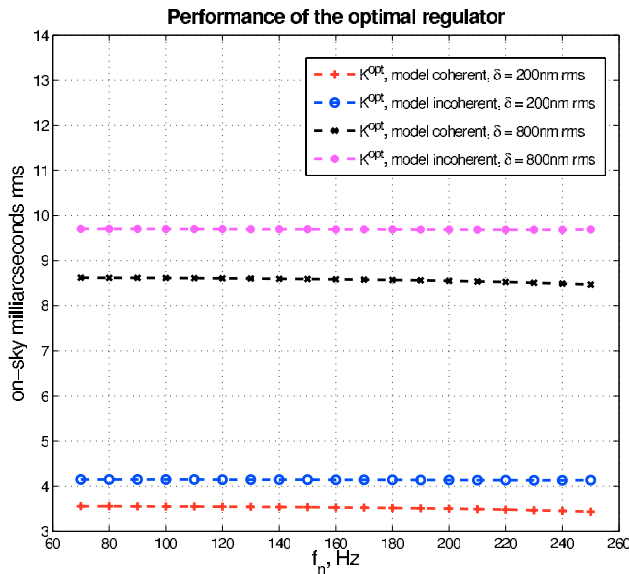


Fig. 12. (Color online) Performance of the optimal regulator for two levels of noise. Modeling errors account for less than ≈ 1 mas rms in both cases. The bottom curve with crosses is the same as the bottom curve in Fig. 10. Ninety seconds of simulation are considered.

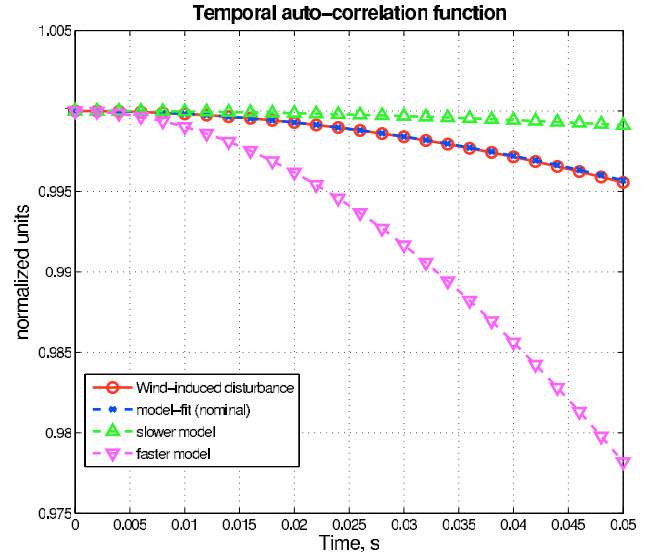


Fig. 13. (Color online) Performance of the optimal regulator with the choice of the disturbance parameters. The bottom curve with crosses is the same as the bottom curve in Fig. 10. Ninety seconds of simulation are considered.

disturbance-related modeling errors to the temporal and estimation errors already accounted for in the previous cases. Monte Carlo results are depicted in Fig. 12 for two noise levels.

Modeling errors are found to account for roughly 0.5 and 1 mas rms in the $\delta=200$ nm rms and $\delta=800$ nm rms noise levels, respectively. Such results indicate that the disturbance model has been correctly identified since the modeling errors' contribution to the overall error budget is small (there is less than 1 on-sky milliarcsec rms additional error).

To further test the choice of disturbance parameters proposed in Subsection 5.A, two other regulators were synthesized based on two distinct disturbance models:

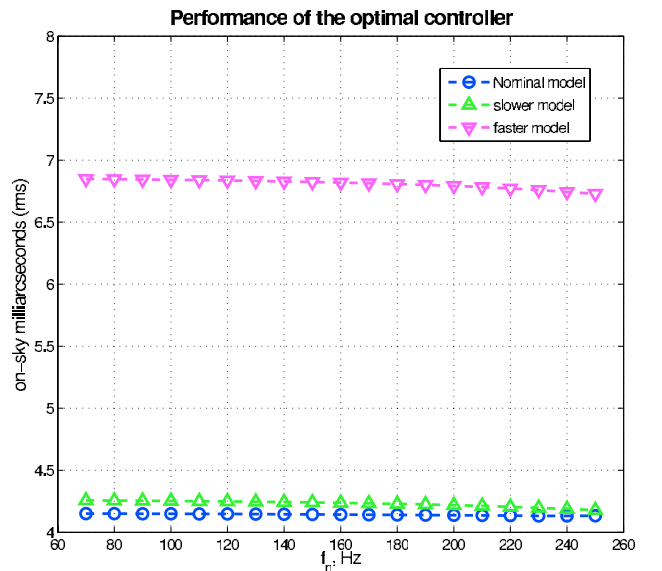


Fig. 14. (Color online) Temporal auto-correlation of the various disturbance models used to derive the regulators: nominal, slower, and faster cases. To be compared with the disturbance auto-correlation of the wind-buffeting series WB.

one whose auto-correlation function decreases *faster* than the nominal and other that decreases *more slowly* as shown in Fig. 13. These models were arbitrarily tuned to have a $5\times$ faster/slower decorrelation rate with respect to the nominal model at $5\times T_s$.

Figure 14 shows the performance results obtained. In both cases whose parameters depart from the nominal ones, an additional error factor is obtained. It is smaller for the slower case, suggesting that modeling the disturbance as a slower evolving process is less damaging than modeling it as a fast evolving one. Thus, the choice of the disturbance parameters that fit the first points of the auto-correlation function of the disturbance proves meaningful since the nominal parameters ensure the smallest residual. Due to the sensitivity to disturbance model parameters, this result further underlines the need to thoroughly know/identify the disturbance's parameters if an accurate control-oriented model is to be defined and used to synthesize the optimal regulator.

6. SUMMARY

This contribution reports on the impact of DM dynamics on optical performance of AO systems. A minimum-variance approach based on the LQG formalism, generalizing the previous infinitely-fast-DM control-oriented model was developed.

An extended state-space model accommodates both the disturbance and DM partial models and allows one to (1) synthesize the optimal MV regulator and (2) assess the performance of both optimal and *any* linear sub-optimal solutions.

LTI DM models of the second order with strongly oscillatory responses and damping factors that result from weakly dissipative structures were considered. Such extreme cases cover a likely range of parameters of present and soon-to-be deformable secondaries and ELTs pre-focal DMs. Second-order disturbance models were identified from European-ELT-like atmospheric phase and wind-buffeting PSD by fitting the first steps of the auto-correlation function in the vicinity of zero.

The results obtained show that neglecting DM dynamics gives rise to worst performance in terms of both residual phase and stability. The DM tends to oscillate with large amplitudes due to noise amplification at the resonant frequency. These results were cross checked analytically and using Monte Carlo end-to-end simulations. Moreover, the choice of disturbance-model parameters has been validated. The chosen model ensures lowest residual when compared with that of models whose parameters depart from the nominal ones.

Monte Carlo simulations with model-independent trajectories have shown that second-order models are rich in grasping to a great extent the disturbance's main features (PSD at low and high frequencies, auto-correlation function). This results in modeling errors that do not exceed 1 mas rms in the overall error budget (for the noise levels considered here). The optimal regulator grants stability for a broad range of natural resonant frequencies and damping coefficients.

An important result is that DM manufacturers' constraints can be relaxed (to a limited extent) since the dy-

namical behavior of the DM can be dealt with on the controller's side, upon correct knowledge of the system parameters. Besides, the optimal controller improves both stability and performance at the same time.

For the time being, complete characterizations of DM dynamics of realistic devices are still rare. Only upon these can one assess whether complex regulators are needed or whether simpler solutions suffice (that ensure high performance and stability). To overcome the lack of knowledge of DM dynamics, one possible solution that may easily be implemented is to install position sensors at least in the TT mirror to supervise model mismatching and on-line model adaptation.

APPENDIX A: CRITERION DISCRETIZATION

Property 2: *Exact discretization formula.*

The mirror deformation $p(t)$ at time $t \in [kT_s, (k+1)T_s[$ is given by [18]

$$p(t) = C_m e^{(t-kT_s)A_m} x_k^m + \left(\int_0^{t-kT_s} C_m e^{vA_m} B_m dv + D_m \right) u_k. \quad (A1)$$

Denote as F the transfer function from u to p , i.e., $F(s) = C_m(sI - A_m)^{-1}B_m + D_m$. From point (c) in Eq. (28), $F(0) = D_m - C_m A_m^{-1} B_m = I$. Using this and the identity $\int_0^s e^{vA_m} dv = A_m^{-1}(e^{sA_m} - I)$, one gets

$$p(t) = C_m e^{(t-kT_s)A_m} x_k^m + (I + C_m e^{(t-kT_s)A_m} A_m^{-1} B_m) u_k, \quad (A2)$$

which is Eq. (30). Then

$$\frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} p(t)^T N^T N p(t) dt = x_k^{m,T} Q_0 x_k^m + 2x_k^{m,T} S_0 u_k + u_k^T R_0 u_k, \quad (A3)$$

$$\frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} p(t)^T N^T \phi^{\text{tur}}(t) dt = u_k^T N^T \bar{\phi}_{k+1}^{\text{tur}} + (x_k^m + A_m^{-1} B_m u_k)^T \bar{\varphi}_{k+1}^{\text{tur}}. \quad (A4)$$

Developing Eq. (2) with the expressions obtained above leads to the discrete criterion

$$J_0^d(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} (z_k^{\text{crit}})^T \bar{Q} z_k^{\text{crit}} + 2(z_k^{\text{crit}})^T \bar{S} u_k + u_k^T \bar{R} u_k, \quad (A5)$$

with

$$\bar{Q} \triangleq \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -I \\ 0 & -I & Q_0 \end{pmatrix}, \quad \bar{S} \triangleq \begin{pmatrix} -N \\ -A_m^{-1} B_m \\ S_0 \end{pmatrix}, \quad \bar{R} \triangleq R_0 > 0, \quad (A6)$$

where Q_0, S_0 and R_0 are defined in Eq. (39). It can readily be checked that

$$P \triangleq \begin{pmatrix} \bar{Q} & \bar{S} \\ \bar{S}^\top & \bar{R} \end{pmatrix} \not\geq 0. \quad (\text{A7})$$

Equivalent discrete-time criterion: The criterion defined by Eq. (A5) is not a quadratic form [matrix P in Eq. (A7) is not semi-positive definite]. To deal with, one generates terms independent of u to be added to the diagonal of the upper-left half-matrix to verify the property. The eventually added terms need be removed from the final computation of the continuous criterion.

Property 3 $\exists \varepsilon > 0$ such that

$$\bar{Q}_1 \triangleq \bar{Q} + \bar{Q}_\varepsilon \triangleq \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -I \\ 0 & -I & Q_0 \end{pmatrix} + \begin{pmatrix} \frac{2}{\varepsilon} NN^\top & 0 & 0 \\ 0 & \frac{1}{\varepsilon} (I + 2B_m^\top A_m^{-1} B_m) & 0 \\ 0 & 0 & 0 \end{pmatrix} \geq 0, \quad (\text{A8})$$

and thus

$$P_1 \triangleq \begin{pmatrix} \bar{Q}_1 & \bar{S} \\ \bar{S}^\top & \bar{R} \end{pmatrix} \geq 0. \quad (\text{A9})$$

With these definitions, the elementary cost in Eq. (2) becomes

$$\begin{aligned} J^c(u)_k &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{res}}(t)\|^2 dt \\ &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t)\|^2 dt + \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}^\top \begin{pmatrix} \bar{Q} & \bar{S} \\ \bar{S}^\top & \bar{R} \end{pmatrix} \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix} \\ &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t)\|^2 dt + J_0 + \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}^\top \begin{pmatrix} \bar{Q}_1 & \bar{S} \\ \bar{S}^\top & \bar{R} \end{pmatrix} \begin{pmatrix} z_k^{\text{crit}} \\ u_k \end{pmatrix}. \end{aligned} \quad (\text{A10})$$

The term J_{tur} in Eq. (31) gathers the first two terms in Eq. (A10) that are independent from u . When Eq. (A10) is plugged in Eq. (3), the first term is the disturbance variance, whereas the second J_0 is the converse of the contribution brought in by the addition of the terms independent of the control u through inclusion of matrix Q_ε in Eq. (A8). This ensures positive definiteness of P_1 in Eq. (A9). By the definition of $J^c(u)_k$ in Eq. (3) one obtains Eq. (A10) [or equivalently Eq. (2)].

Proof 1: Since Eq. (A3) is the integral of a non-negative function, so is the associated quadratic form:

$$P_0 \triangleq \begin{pmatrix} Q_0 & S_0 \\ S_0^\top & R_0 \end{pmatrix} \geq 0. \quad (\text{A11})$$

Suppose that Eq. (A3) is zero for some (x_k, u_k) . Since $N^\top N$ is invertible and p is a continuous function over $[kT_s, (k+1)T_s]$, it follows that $p(t)$ is everywhere zero on this interval and thus that all its right-derivatives at $t = kT_s$ are also zero:

$$p_k = p(kT_s) = C_m x_k^m + D_m u_k = 0, \quad (\text{A12})$$

$$\forall i > 0, \quad \frac{d^i p}{dt^i}(kT_s) = C_m A_m^i x_k^m + C_m A_m^{i-1} B_m u_k = 0. \quad (\text{A13})$$

This last identity means that for all $i \geq 0$, $A_m x_k^m + B_m u_k \in \ker(C_m A_m^i)$, in other words that $A_m x_k^m + B_m u_k$ is an unobservable initial condition for (A_m, C_m) . Since (A_m, C_m) is observable, $A_m x_k^m + B_m u_k = 0$, so that (x_k^m, u_k) is an equilibrium point of Eq. (28). Because the DC gain between u and p is unitary, this equilibrium point verifies that $C_m x_k^m + D_m u_k = 0$ and $u_k = 0$. It follows that x_k^m is also an unobservable initial condition for (A_m, C_m) and thus that $(x_k^m, u_k) = 0$. This establishes that $P_0 > 0$.

Finally, because $P_0 > 0$, one can find $\varepsilon > 0$ such that $P_0 > \varepsilon I$. Using Schur complements and elementary matrix manipulations, it is easily checked that this guarantees $P_1 \geq 0$, which completes the proof. ■

APPENDIX B: CONSTRUCTION OF THE DISCRETE-TIME STATE SPACE

Property 4: Exact discretization of continuous-time state-space models.

Consider the general continuous stochastic state-space system, with v a Gaussian white noise

$$\dot{x}(t) = A_c x(t) + B_c u(t) + v(t). \quad (\text{B1})$$

The exact discretization of the system (B1) is given by

$$x_{k+1} = A_d x_k + B_d u_k + v_k, \quad (\text{B2})$$

with

$$A_d \triangleq e^{A_c T_s}, \quad B_d \triangleq \int_0^{T_s} e^{A_c B_c} dt, \quad (\text{B3})$$

and noise v_k with covariance

$$\Sigma_v \triangleq \int_0^{kT_s} e^{t A_c} \Sigma_v e^{t A_c^\top} dt. \quad (\text{B4})$$

Solution of the differential equation $\dot{r}_{\text{tur}}(t) = 1/T_s \phi^{\text{tur}}$ is $r_{\text{tur}}(t) = 1/T_s \int_0^t \phi^{\text{tur}}(s) ds = C_{\text{tur}}/T_s \int_0^t x^{\text{tur}}(s) ds$. Let the continuous-time state-space system

$$\begin{pmatrix} \dot{x}^{\text{tur}} \\ \dot{r}^{\text{tur}} \end{pmatrix} = \begin{pmatrix} A_{\text{tur}} & 0 \\ C_{\text{tur}} & 0 \end{pmatrix} \begin{pmatrix} x^{\text{tur}} \\ r^{\text{tur}} \end{pmatrix} + \begin{pmatrix} \eta \\ 0 \end{pmatrix}, \quad (\text{B5})$$

where $\eta(t)$ is the state noise with covariance matrix Σ_η . The first line defines Eq. (40). Using property 4 for the exact discretization of state-space systems, after some manipulation, it can be found that Eq. (B5) becomes in finite differences

$$\begin{pmatrix} x_{k+1}^{\text{tur}} \\ r_{k+1}^{\text{tur}} \end{pmatrix} = \begin{pmatrix} e^{T_s \mathcal{A}^{\text{tur}}} & 0 \\ \frac{C_{\text{tur}}}{T_s} (e^{T_s \mathcal{A}^{\text{tur}}} - I) & I \end{pmatrix} \begin{pmatrix} x_k^{\text{tur}} \\ r_k^{\text{tur}} \end{pmatrix} + \begin{pmatrix} v_k \\ v_k^r \end{pmatrix}. \quad (\text{B6})$$

Note that the second line produces integrals of the form $1/T_s \int_{kT_s}^{(k+1)T_s} \phi(t) dt + 1/T_s \int_0^{kT_s} \phi(t) dt$, the solution of the differential equation stated above. To retain the slab of the integral over $t \in [kT_s, (k+1)T_s]$, by identification

$$\begin{pmatrix} x_{k+1}^{\text{tur}} \\ \phi_{k+1}^{\text{tur}} \end{pmatrix} = \begin{pmatrix} e^{T_s \mathcal{A}^{\text{tur}}} & 0 \\ \frac{C_{\text{tur}}}{T_s} (e^{T_s \mathcal{A}^{\text{tur}}} - I) & 0 \end{pmatrix} \begin{pmatrix} x_k^{\text{tur}} \\ \phi_k^{\text{tur}} \end{pmatrix} + \begin{pmatrix} v_k \\ v_k^r \end{pmatrix}. \quad (\text{B7})$$

Proceeding in the same manner to obtain the DM-weighted phase, define

$$\begin{pmatrix} \dot{x}^{\text{tur}} \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} \mathcal{A}^{\text{tur}} & 0 \\ \frac{1}{T_s} C_m^T N^T C_{\text{tur}}^T & -\mathcal{A}_m^T \end{pmatrix} \begin{pmatrix} x^{\text{tur}} \\ x_2 \end{pmatrix} + \begin{pmatrix} \eta \\ 0 \end{pmatrix}. \quad (\text{B8})$$

Using developments analogous to those above, one gets integrals of the form $1/T_s \int_{kT_s}^{(k+1)T_s} e^{-\mathcal{A}_m t} \phi(t) dt$. Gathering all, applying property 4, matrices Ξ and Φ are determined from the exact discretization of the continuous-time state-space system with transition matrix

$$\mathcal{A}_c \triangleq \begin{pmatrix} \mathcal{A}^{\text{tur}} & 0 & 0 \\ \frac{1}{T_s} C_{\text{tur}}^T & 0 & 0 \\ \frac{1}{T_s} e^{T_s \mathcal{A}_m^T} C_m^T N^T C_{\text{tur}}^T & 0 & -\mathcal{A}_m^T \end{pmatrix}. \quad (\text{B9})$$

Therefore

$$\Xi \triangleq (I \ 0 \ 0) e^{T_s \mathcal{A}_c} (0 \ I \ 0)^T, \quad (\text{B10})$$

$$\Phi \triangleq (I \ 0 \ 0) e^{T_s \mathcal{A}_c} (0 \ 0 \ I)^T. \quad (\text{B11})$$

Finally,

$$\Sigma_v \triangleq \int_0^{T_s} e^{s \mathcal{A}_c} (I \ 0 \ 0) \Sigma_\eta (I \ 0 \ 0)^T e^{s \mathcal{A}_c^T} ds. \quad (\text{B12})$$

APPENDIX C: DEFORMABLE MIRROR MODAL MODEL

Equation (27) defines a very general class of the DM's dynamics. One important special case is when the DM's response can be decomposed according to a set of independent mechanical eigenmodes. Assuming that the total number of eigenmodes retained in the model is n_e , this would result in a relation in the form

$$\phi^{\text{cor}}(t) = \sum_{j=1}^{n_e} v_j d_j(t), \quad (\text{C1})$$

where v_j is the j th DM-surface eigenvector, and the scalar function $d_j(t)$ its displacement at time t . To fit the assumptions in hypothesis 1, it is necessary to assume that the subspace spanned by these surface eigenvectors is

contained into the subspace spanned by the static DM's influence functions. Thus, without loss of generality, one may assume that $n_e = \dim(u)$, and that $V \triangleq (v_1 \cdots v_{n_e})$ has full column rank $\dim(u)$. The identities $Vd = Np$ and $Ve = Nu$, taken together, define a bijection between the input-output pairs (u, p) and (e, d) . Assume now that the input-output model for each DM eigenmode is a simple second-order temporal response with unitary DC gain, natural frequency ω_j , and damping coefficient ξ_j :

$$\ddot{d}_j(t) + 2\xi_j \omega_j \dot{d}_j(t) + \omega_j^2 d_j(t) = \omega_j^2 e_j(t). \quad (\text{C2})$$

Then a simple state-space model for the dynamics between e_j and d_j is therefore

$$\dot{\vartheta}(t) = \begin{pmatrix} \dot{\vartheta}_1(t) \\ \dot{\vartheta}_2(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega_j^2 & -2\xi_j \omega_j \end{pmatrix} \begin{pmatrix} \vartheta_1(t) \\ \vartheta_2(t) \end{pmatrix} + \begin{pmatrix} 0 \\ \omega_j^2 \end{pmatrix} e_j(t), \quad (\text{C3})$$

$$d_j(t) = (1 \ 0) \begin{pmatrix} \vartheta_1(t) \\ \vartheta_2(t) \end{pmatrix}. \quad (\text{C4})$$

Defining Ω as the diagonal matrix with general term ω_j and Λ the diagonal matrix with general term ξ_j , this set of decoupled models can be concatenated into a single state-space model with input e , output d , and state vector x_m :

$$\dot{x}_m(t) = \begin{pmatrix} 0 & I \\ -\Omega^2 & -2\Omega\Lambda \end{pmatrix} x_m(t) + \begin{pmatrix} 0 \\ \Omega^2 \end{pmatrix} e(t), \quad (\text{C5})$$

$$d(t) = (I \ 0) x_m(t). \quad (\text{C6})$$

For the system with input u and output p , this corresponds to a state-space model in the form of Eq. (28), with

$$\mathcal{A}_m = \begin{pmatrix} 0 & I \\ -\Omega^2 & -2\Omega\Lambda \end{pmatrix}, \quad \mathcal{B}_m = \begin{pmatrix} 0 \\ \Omega^2 \end{pmatrix} (V^T V)^{-1} V^T N, \quad (\text{C7})$$

$$C_m = (N^T N)^{-1} N^T V (I \ 0), \quad \mathcal{D}_m = 0. \quad (\text{C8})$$

A useful variant of this construction is derived by assuming that the DM's eigenmodes can be separated into slow modes with second-order dynamics and fast modes with instantaneous response. Let n_s, n_f be the numbers of slow and fast modes, let V be partitioned accordingly as $V = (V_s \ V_f)$, and let Ω_s and Λ_s be the diagonal matrices containing the slow modes' natural frequencies and damping coefficients. With this notation, it is immediately verified that the DM's state space model becomes

$$\begin{aligned} \mathcal{A}_m &= \begin{pmatrix} 0 & I \\ -\Omega_s^2 & -2\Omega_s \Lambda_s \end{pmatrix}, \\ \mathcal{B}_m &= \begin{pmatrix} 0 \\ \Omega_s^2 \end{pmatrix} (I \ 0) (V^T V)^{-1} V^T N, \\ C_m &= (N^T N)^{-1} N^T V \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}, \\ \mathcal{D}_m &= (N^T N)^{-1} N^T V \begin{pmatrix} 0 \\ I \end{pmatrix}. \end{aligned} \quad (\text{C9})$$

Note that the application to tip/tilt mirror control presented in Section 5 can be seen as a prototypical simple case of such eigenmode-based models.

ACKNOWLEDGMENTS

The authors acknowledge fruitful discussions with Dominique Le Bihan at ONERA and Rami Gasmi at Observatoire Paris-Meudon on the DM dynamical models. We also acknowledge the European Southern Observatory in the person of B. Sedghi for providing the wind-buffeting data. Funding was partially provided by Fundação para a Ciência e a Tecnologia under contract number SFRH/BD/30010/2006. This work also received the support of PHASE, the high angular resolution partnership between ONERA, Observatoire de Paris, CNRS, and Université Paris Diderot.

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